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RESEARCH INTERESTS

- Autonomous condition monitoring-based asset management for transportation infrastructure
- Optimization of planning and operations
- Transportation data analytics and artificial intelligence
- Advanced air mobility systems

EDUCATION

- Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea
 - Ph.D. in the Graduate School of Mobility, 2023
 - Dissertation title: Optimal road asset management system considering condition inspection policies and information value
- Korea University, Seoul, Korea
 - M.S. in Civil and Environmental Engineering, 2019
 - Thesis title: New section based congestion index for applications in enhancing missing value imputation and predicting travel time
 - B.S. in Civil and Environmental Engineering, 2017

ACADEMIC EXPERIENCE

- KAIST, Daejeon, Korea
 - Researcher, 2019-2020
 - Postdoctoral Researcher, 2023
- UIC, Illinois, United States
 - Visiting Scholar, 2023-2024
 - Postdoctoral Researcher, 2024-2025

PEER-REVIEWED JOURNAL PUBLICATIONS AND UNDER REVIEW

Transportation Asset Management Systems

- **Shon, H.** and Lee, J., 2021. Integrating multi-scale inspection, maintenance, rehabilitation, and reconstruction decisions into system-level pavement management systems. *Transportation Research Part C: Emerging Technologies*, 131, p.103328.

- **Shon, H.**, Cho, C., Byon, Y. and Lee, J., 2022. Autonomous condition monitoring-based pavement management system (ACM-PMS). *Automation in Construction*, 138, p.104222.

Advanced Air Mobility Systems

- **Shon, H.**, Kim, S., and Lee, J., 2024. Optimal planning of urban air mobility systems accounting for ground access trips. *International Journal of Sustainable Transportation*, 18(4), pp.356-378.
- **Shon, H.** and Lee, J., 2025. An optimization framework for urban air mobility (UAM) planning and operations. *Journal of Air Transport Management*, 124, p.102720.
- Perez, D., **Shon, H.**, Zou, B. and Kuhn, K., 2025. Advanced air mobility for commuting? An exploration of economic, energy, and environmental feasibility. *Transport Economics and Management*, 3, pp.135-152.

Big Data Analysis and Deep Learning

- Choi, Y.Y., **Shon, H.**, Byon, Y.J., Kim, D.K. and Kang, S., 2019. Enhanced application of principal component analysis in machine learning for imputation of missing traffic data. *Applied Sciences*, 9(10), p.2149.
- **Shon, H.**, Kim, S., Yoon, Y. and Lee, J. Incorporating Traffic Flow Transmission Impacts into Dynamic Graph Diffusion Convolutional Network for Traffic Speed Prediction. *Under review*.

Transportation Planning and Operations

- Lee, J., **Shon, H.**, Papakonstantinou, I. and Son, S., 2021. Optimal fleet, battery, and charging infrastructure planning for reliable electric bus operations. *Transportation Research Part D: Transport and Environment*, 100, p.103066.
- Jeong, M., **Shon, H.**, Kim, J., Jung, Y. and Lee, J., 2023. Analysis of traffic load reduction effects of overloading enforcement system from a traffic assignment perspective, *International Journal of Highway Engineering*, 25(6), pp.309-315.
- Jeong, M., **Shon, H.**, Kim, J., Jung, Y. and Lee, J. Developing Link Travel Time Functions for Highway Merging and Diverging Segments with Mixed Truck Traffic. *Under review*.

PEER-REVIEWED CONFERENCE PROCEEDINGS

- **Shon, H.** and Lee, J. 2022. Optimal Urban Air Mobility (UAM) Planning and Operations. *Transportation Research Board (TRB) 102nd Annual Meeting*. Accepted for presentation.
- **Shon, H.**, Kim, S. and Lee, J., 2022. Incorporating Traffic Flow Transmission Impacts into Spatio-Temporal Graph Diffusion Convolutional Network for Traffic Speed Prediction. *Transportation Research Board (TRB) 102nd Annual Meeting*. Accepted for presentation.

- Selmoune, A., **Shon, H.**, Han, J., Kim, J. and Lee, J., 2022. Heterogeneity in Bus Choice Behavior by Time of Day: Empirical Evidence from Smart Card Big Data. *Transportation Research Board (TRB) 102nd Annual Meeting*. Accepted for presentation.
- **Shon, H.**, Cho, C., Byon, Y. and Lee, J. *, 2022. Initial Concept of Autonomously Monitored Pavement Management Systems and Its Potential Benefits. *Transportation Research Board 101st Annual Meeting*. Accepted for presentation.
- **Shon, H.** and Lee, J. *, 2021. Information Value in Pavement Management Systems with Surface Condition Information Collected by Autonomous Vehicles. *The 14th International Conference of Eastern Asia Society for Transportation Studies (EASTS)*.
- **Shon, H.** and Lee, J. *, 2021. Joint optimization of multi-scale decisions in budget allocation, inspection frequency, and maintenance policies for transportation infrastructure systems. *Transportation Research Board 100th Annual Meeting*. Accepted for presentation.
- Oh, Y., **Shon, H.**, Jeong, H., Bae, HG. and Kang, S. *, 2018. Analysis on taxi operation using digital tachograph and meter data in Korea. *18th International Conference Road Safety on Five Continents (RS5C 2018)*.

HONORS AND AWARDS

- Best paper award, The 86th Conference of Korean Society of Transportation, 2022
 - Presentation title: Bus Network Optimization Using Multi-Agent Reinforcement Learning
- Best paper award, Conference of Korean Society of Civil Engineering, 2021
 - Presentation title: Initial Concept of Autonomously Monitored Pavement Management System and Potential Benefits
- Best paper award, The 83rd Conference of Korean Society of Transportation, 2020
 - Presentation title: Joint Optimization of Pavement Condition Inspection and Maintenance Strategies
- President award, Korean Society of Transportation, 2017

SELECTED RESEARCH & APPLIED PROJECTS

- Preliminary Study for a Road Construction Project, *Korea Development Institute*
 - Built EMME-based multimodal network scenarios and performed traffic assignment to quantify travel-time savings, safety, and emissions as inputs to cost-benefit analysis. (*Methods/Tools: EMME, Python, demand-supply scenarios*)
- Taxi Operation Information Management System, *Neighbor System*
 - Analyzed large-scale digital tachograph (DTG) taxi data to identify demand hotspots and prototyped detectors for regulatory violations/unauthorized operations. (*Methods/Tools: Statistics, Python, GIS*)

- Road Asset Management System (RAMS) Strategy for the Dominican Republic, *The Export-Import Bank of Korea*
 - Designed a country-tailored RAMS framework (master plan, data collection and pavement-condition indices, organizational roles, prioritization/repair decision rules, system architecture), and produced a phased implementation roadmap.
- Big-Data- and Optimization-Driven Sustainable Transport Infrastructure Asset Management, *National Research Foundation of Korea*
 - Designed multi-objective life-cycle maintenance optimization and a connected-vehicle-based condition-monitoring pipeline (*Results: two papers in top-tier journals-Transportation Research Part C; Automation in Construction.*)
- Urban Mobility Network Analysis & Forecasting with Attention Mechanisms and Graph Neural Networks, *National Research Foundation of Korea*
 - Built a graph-based neural model with attention and diffusion to learn taxi-flow spatial correlations and forecast speeds from DTG data, with map matching and a recurrent encoder. (*Methods/Tools: graph diffusion convolutional networks, recurrent neural network, DTG data, Python*)
- Jeju Innovation City Smart-Mobility Living Lab, *Jeju Free International City Development Center*
 - Developed a prototype where roadside video infrastructure detects difficult perception cases and sends hazard alerts to autonomous vehicles; implemented end-to-end code from camera ingestion to alerting. (*Methods/Tools: autonomous vehicles, convolutional neural networks, Python, video analytics*)

TEACHING EXPERIENCE

- Research assistant, Korea University - ACE Capstone Design (Spring 2017)
 - Mentored undergraduate project teams on topic selection and research design
 - Provided technical guidance on traffic simulation modeling and experiment setup
 - Reviewed reports and presentations, giving feedback on methods, analysis, and clarity
- Teaching assistant, Korea University - Surveying (Fall 2017)
 - Organized and led surveying labs and field exercises (leveling, total station)
 - Graded quizzes, lab reports, and exams; provided timely feedback
 - Managed equipment checkout and calibration; held office hours to support students

LANGUAGES

- English (Proficient)
- Korean (Native)

SKILLS

Proficient Python, Matlab, R, EMME, and Microsoft Office

Working C++, VISSIM, CORSIM, and LaTeX

MILITARY SERVICE

- Sergeant, The Republic of Korea's Army, Korea